Explainable AI for Heart Sound Analysis: Bridging the Gap Between Algorithmic Inference and Clinical Reasoning

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Project guide

***Abstract*—Accuracies reported on the PhysioNet/CinC 2016 heart sound benchmark rarely survive a change of recording site. This paper audits an explainable ResNet50 classifier with squeeze-and-excitation (SE) and multi-head self-attention (MHSA) modules through five measurements: duplicate leakage, a three-seed component ablation, leave-one-database-out (LODO) testing, a population-level explanation audit and calibration at a sensitivity-targeted threshold. The 301 validation-folder files duplicate training recordings. Removing them leaves 3,240 unique recordings and leaves test accuracy unchanged (86.7% against 86.6%). Across three seeds, ResNet50 with SE reaches 89.0% balanced accuracy and 95.4% AUC-ROC with 25.1 M parameters. Adding MHSA raises the count to 41.9 M without a measurable gain. LODO testing lowers mean balanced accuracy to 52.7% for SE+MHSA and 49.1% for the plain backbone. Occluding 20 to 150 Hz lowers balanced accuracy by 7.8 points, more than random controls do and Grad-CAM passes a cascading randomization check. However, deletion tests show no attribution method is more faithful than random removal and the methods agree weakly. These results place cross-site generalization ahead of architecture choice as the main obstacle and call for patient-disjoint multi-site validation.**

***Index Terms*—phonocardiogram, heart sound classification, data leakage, component ablation, explainable AI, model calibration, domain shift**

I. Introduction

Cardiovascular disease caused an estimated 17.9 million deaths in 2019, representing 32% of global mortality, with 85% attributable to heart attacks and strokes [1]. Frontline screening relies on auscultation, where examiners listen for the first and second heart sounds S1 (25 to 100 Hz) and S2 (50 to 150 Hz) and for murmurs (150 to 500 Hz). The method is subjective and vulnerable to ambient noise, respiration and sensor contact artifacts.

Automated analysis of phonocardiograms (PCG) offers an objective pre-screening tool in which sensitivity matters most, because a missed pathology costs more than a false alarm. Deep networks score highly on PhysioNet/CinC 2016 [19], yet three problems weaken those scores. Random splits can place recordings from one site or patient on both sides of the split. The corpus contains duplicate files. Explanation studies often show saliency maps for a few chosen recordings without population-level checks, although single saliency maps can mislead [7].

This paper examines the architecture of [5] on unsegmented recordings through four contributions:

1) **Leakage-aware benchmark.** Duplicates are isolated and results are compared against the legacy split. Leave-one-database-out testing supplies a cross-site evaluation (Sections V-A and V-C).
2) **Component ablation** of the ResNet50, SE and MHSA modules over three seeds, plus a dilated backbone with a 14×14 grid (Section V-B).
3) **Population-level explanation audit** covering frequency occlusion with random controls, deletion faithfulness, cascading randomization and multi-method agreement (Section V-D).
4) **Calibration and operating point.** Expected calibration error and a validation-selected threshold targeting sensitivity of at least 0.90 (Section V-E).

II. Literature Review

Early PCG classifiers combined hand-built time-frequency features, such as wavelet coefficients and mel-frequency cepstral coefficients, with hidden Markov models or support vector machines. Rubin et al. [2] trained a two-stage CNN on spectrograms and MFCCs for the 2016 challenge. Thomae and Dominik [3] combined a convolutional front end with gated recurrent units. Li et al. [13] used a multi-scale CNN with global average pooling and reported 86.80% accuracy on PhysioNet 2016 without explanations.

Ren et al. [14] added attention pooling to a CNN and evaluated it on the three-class Heart Sounds Shenzhen set, reporting an unweighted average recall of 51.2%. Their attention weights run along time only and give no frequency localization. Padhy et al. [15] combined a CNN with a bidirectional LSTM on log-mel spectrograms and reported 99.15% on a balanced five-class subset of 2,400 recordings. Althaph and Challa [16] proposed a heart sound transformer with 96.7% on HeartWave and 90.3% on PhysioNet 2016. Alrabie and Barnawi [5] paired ResNet50 with SE and multi-head attention on the four-class HeartWave data, reached 97.3% accuracy and scored

Grad-CAM against cardiologist annotations (mean IoU 82%). That scoring needs annotated segment boundaries. Other work applies LIME and SHAP to classical models [17] and to tabular risk data [18]. Table I summarizes these studies.

Evaluation protocols differ widely. Many studies train and test on the full corpus without removing duplicates or split by random recording instead of by site, which permits leakage between recordings from one clinical environment. Methods that need cycle segmentation depend on external tools [4]. The architecture in [5] works on unsegmented 8-second clips and avoids that dependency.

III. Dataset

The PhysioNet/CinC Challenge 2016 corpus [19], [20] collects heart sounds from six independent sources, released as training-a through training-f. Recordings come from healthy volunteers and patients with valvular defects or coronary artery disease, captured with different stethoscopes and microphones in uncontrolled clinical settings. Lengths range from 5 to 120 s. Each recording has one binary label, mapped to 0 for normal and 1 for abnormal.

The working table has 3,541 entries: 3,240 recordings from training-a to training-f plus 301 entries from the challenge validation directory. All 301 validation entries repeat the waveform and label of a training recording. After deduplication the benchmark holds 3,240 unique recordings, 2,575 normal and 665 abnormal (20.5% abnormal). The legacy table has a 77:23 ratio. Class proportions differ sharply by source: training-e is about 91% normal and training-a is mostly abnormal. Subject identifiers are not provided, so recordings from one patient may fall in several partitions.

Three protocols are used. The **legacy split** uses all 3,541 entries (70/15/15, seed 42) and measures what the duplicates change. The **deduplicated benchmark** uses a stratified 70/15/15 split of the 3,240 recordings with seeds 42 to 44 (Table II); each seed changes both the partition and the initialization. **LODO** holds out training-a, training-b, training-e or training-f in turn and trains on the rest. Subsets c and d have too few recordings to hold out and stay in the training pool.

The challenge scored systems by the mean of sensitivity and specificity, which equals balanced accuracy. Plain accuracy rewards majority-class predictions under the class imbalance, so balanced accuracy is the primary metric here.

IV. Proposed Methodology

A. Experimental Setup

Preprocessing. Each recording is resampled to 4 kHz, which covers energy up to 1,000 Hz without aliasing. A 4th-order Butterworth band-pass filter (25 to 900 Hz) runs forward and backward for zero phase distortion, which keeps S1 and S2 aligned in time. It removes respiratory rumble below 25 Hz and transducer noise above 900 Hz. Signals are peak-normalized to $[-1, 1]$ and padded with zeros or cropped from the onset to 8 s (32,000 samples). An 8 s window holds 8 to 13 cardiac cycles at resting heart rates.

Features. A 512-sample Hann window (128 ms) with a hop of 128 samples (32 ms) feeds 128 Slaney mel filterbanks from 20 to 1,000 Hz. Below 1,000 Hz the Slaney scale is linear, $f = \frac{200}{3} m$, which gives roughly 7.7 Hz between adjacent bands. Power is converted to decibels relative to the peak and scaled to $[0, 1]$, giving a 128×251 matrix. For the network the matrix is resized bilinearly to 224×224, replicated across three channels and normalized with ImageNet statistics [22].

Augmentation. SpecAugment [21] is applied in log-mel space before resizing: two frequency masks (1 to 17 bands) and two time masks (1 to 24 frames), set to zero.

Training. The network starts from ImageNet weights. In Phase A (10 epochs) the backbone is frozen, its batch-normalization statistics are held in evaluation mode; the SE block, attention layer and classifier train with AdamW [23] at 3×10^{-4} under cosine annealing [24]. In Phase B (30 epochs) all layers train at 3×10^{-5} decaying to 10^{-7}. Early stopping (patience 8 on validation loss) restarts at the beginning of Phase B and the lowest-validation-loss checkpoint is kept. The loss is class-weighted cross-entropy with label smoothing $\alpha = 0.05$ [25]:

$$\mathcal{L} = -\sum_{c=0}^{1} w_c \, q_c \log p_c, \quad q_c = (1 - \alpha)\, y_c + \frac{\alpha}{2}, \tag{1}$$

with $w_c = 1 - N_c/N$ computed on the training split (about 0.21 for normal and 0.79 for abnormal). Other settings: batch size 32, weight decay 10^{-4}, gradient clipping at norm 1.0, head dropout 0.4 and 0.2, attention dropout 0.1 and mixed precision [26] on one NVIDIA T4 GPU.

Metrics. Abnormal is the positive class. We report accuracy, balanced accuracy, macro precision, recall and F1, AUC-ROC, AUC-PR and Wilson 95% intervals for proportions.

B. Models

Backbone. A ResNet50 [8] without its pooling and classification layers maps the input to a $2048 \times 7 \times 7$ map (23.5 M parameters).

Squeeze-and-excitation. The SE block [9] pools each channel to a descriptor z_c and learns channel weights through a bottleneck with reduction $r = 16$:

$$\mathbf{s} = \sigma\big(\mathbf{W}_2 \, \delta(\mathbf{W}_1 \mathbf{z})\big), \qquad \tilde{F}_c = s_c F_c, \tag{2}$$

where δ is ReLU and σ is the sigmoid.

Self-attention. The map is flattened into 49 tokens of dimension 2048. Eight-head self-attention [10] with $d_k = 256$ computes

$$\mathrm{head}_i = \mathrm{softmax}\left(\frac{Q_i K_i^{\top}}{\sqrt{d_k}}\right) V_i, \tag{3}$$

followed by a residual connection and LayerNorm. Each cell spans about 1.14 s, longer than one cardiac cycle, so the layer relates time-frequency regions of the clip and does not model single beats.

Classifier. Tokens are averaged and passed through dropout, a 512-unit layer with ReLU, dropout and a two-unit output.

TABLE I
SUMMARY OF RELATED WORK AND THIS STUDY (N/R: NOT REPORTED; THE 2016 CHALLENGE ENTRIES REPORT THE MEAN OF SENSITIVITY AND SPECIFICITY)

Study	Dataset	Classes	Architecture	Accuracy (%)	AUC	Explanation
Rubin et al. [2] (2016)	PhysioNet 2016	2	Two-stage CNN on MFCCs	Score 84.0	N/R	Heatmaps
Thomae and Dominik [3] (2016)	PhysioNet 2016	2	CNN with gated RNN	Score 86.0	N/R	None
Li et al. [13] (2020)	PhysioNet 2016	2	Multi-scale CNN, GAP	86.80	N/R	None
Ren et al. [14] (2022)	HSS (not PhysioNet)	3	CNN, temporal attention	UAR 51.2	N/R	Frame attention
Padhy et al. [15] (2025)	PhysioNet subset (2,400 balanced)	5	CNN and BiLSTM	99.15	0.99	Saliency
Althaph and Challa [16] (2025)	HeartWave; PhysioNet 2016	9; 2	Heart sound transformer	96.7; 90.3	N/R	Attention
Alrabie and Barnawi [5] (2025)	HeartWave (segmented)	4	ResNet50, SE, MHA	97.3	N/R	Grad-CAM (mIoU 82%)
This work (SE, 3 seeds)	PhysioNet 2016 (deduplicated)	2	ResNet50, SE	88.5 (BA 89.0)	0.954	Grad-CAM, attention, SHAP

TABLE II
DEDUPLICATED DATASET PARTITION AND CLASS BALANCE (70/15/15, SEED 42)

Partition	**Total**	**Normal**	**Abnormal**	**Abnormal (%)**
Training	2,268	1,802	466	20.5%
Validation	486	386	100	20.6%
Test	486	387	99	20.4%
Total Benchmark	**3,240**	**2,575**	**665**	**20.5%**

TABLE III
IMPACT OF DATA LEAKAGE: LEGACY VS. DEDUPLICATED EVALUATION

Protocol	**Accuracy**	**Balanced Acc.**	**Macro F1**	**AUC**
Legacy Split (with 301 duplicates)	86.7%	87.6%	83.2%	94.
Deduplicated Benchmark (mean)	86.6%	88.3%	82.2%	95.
Legacy Test Sub-Cohort Breakdown:				
Twin rows in train (n=70)		75.7% accuracy		
Unique rows (n=462)		88.3% accuracy		

Variants. Five variants isolate each module: (1) baseline ResNet50, (2) ResNet50+SE, (3) ResNet50+MHSA, (4) ResNet50+SE+MHSA (41,871,938 parameters, the reference attention model) and (5) SE+MHSA with a dilated backbone. Parameter counts are in Table IV. In the dilated variant, `layer4` replaces its stride with atrous convolution [28] (`replace_stride_with_dilation=[False, False, True]`). This yields a 14×14 grid of 196 tokens at 0.57 s per cell with the same parameter count.

C. Explanation Methods and Audit

Methods. Grad-CAM [11] hooks `layer4[-1]`. For predicted class c, activation maps A^k are weighted by pooled gradients and rectified:

$$\alpha_k^c = \frac{1}{Z}\sum_{i,j}\frac{\partial y^c}{\partial A_{ij}^k}, \qquad L^c = \mathrm{ReLU}\Big(\sum_k \alpha_k^c A^k\Big). \tag{4}$$

Attention maps average the self-attention weights over heads and queries. They are class-agnostic. *Gradient SHAP* [12] approximates Shapley values by expected gradients over 32 training spectrograms as background.

Audit. Four analyses test whether the explanations are trustworthy.

1) **Frequency occlusion.** Bands B1 (20 to 150 Hz), B2 (150 to 500 Hz) and B3 (500 to 1,000 Hz) are masked. Each drop in balanced accuracy is compared with 20 random contiguous masks of the same row height. A band counts as relied upon only if its drop exceeds $\mu_{\text{rnd}} - 2\sigma_{\text{rnd}}$.
2) **Deletion and insertion faithfulness.** Pixels are removed or inserted in order of attribution in 5% steps up to 50% [29], with a zero-decibel baseline. Normalized AUC is compared with random attribution through one-sided Wilcoxon signed-rank tests (H_1: the method deletes faster than random).
3) **Cascading randomization.** Weights are reinitialized from the classifier down to early layers and the Spearman correlation with the original Grad-CAM map is tracked [6].
4) **Method agreement.** Grad-CAM, attention and SHAP maps are pooled to 7×7 and compared by Spearman correlation and top-20% IoU.

D. Calibration and Operating Point

Expected calibration error (ECE) uses 15 equal-width bins [27]. The decision threshold is the highest value giving sensitivity of at least 0.90 on validation probabilities. It is then applied once to the test set.

V. RESULTS AND DISCUSSION

A. Data Leakage Audit

Table III compares the legacy and deduplicated splits.

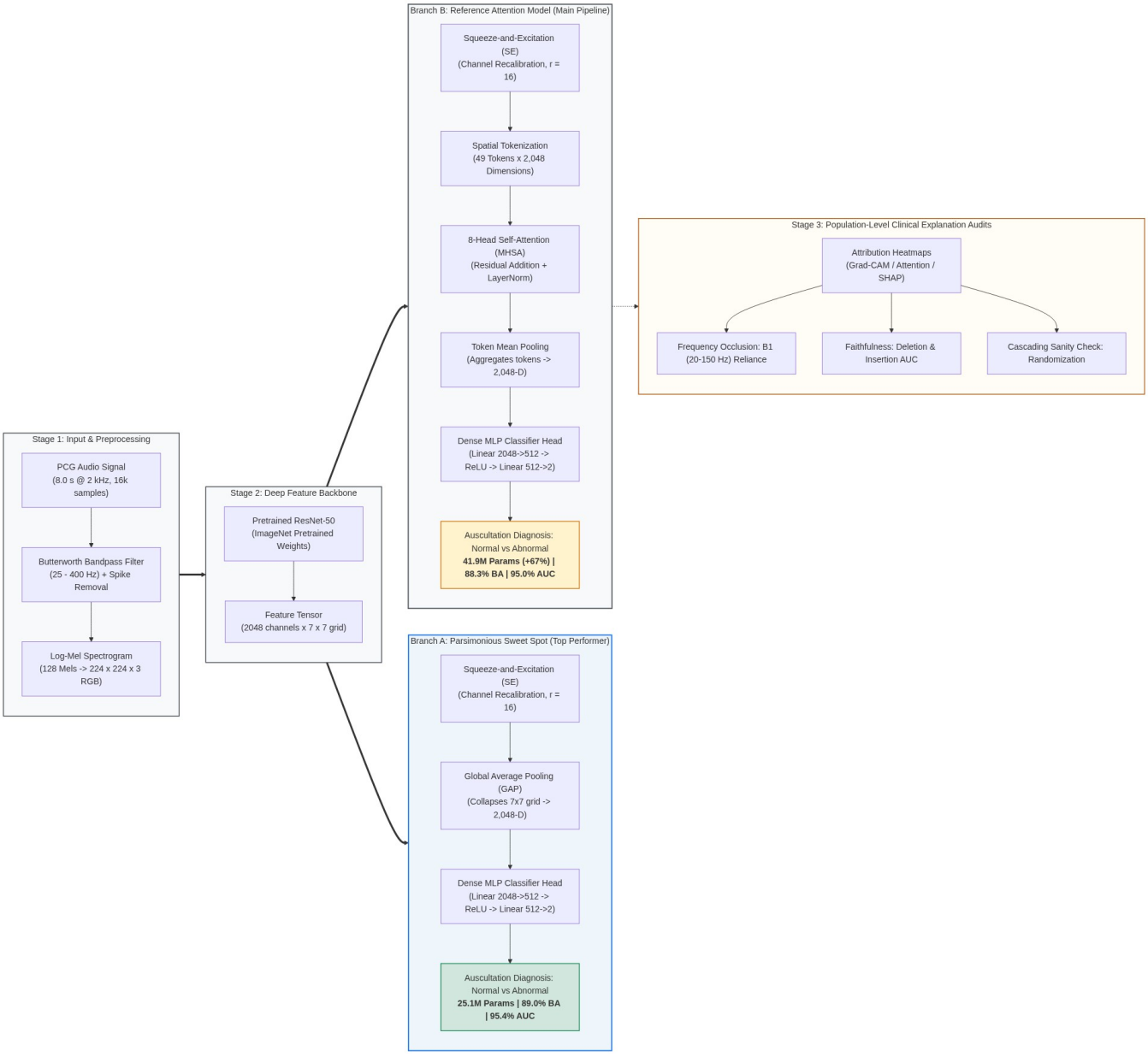


Fig. 1. Architectural comparison. Raw 8-second PCG audio becomes a log-mel spectrogram ($224 \times 224 \times 3$) and passes through a shared ResNet50 backbone ($2048 \times 7 \times 7$). Branch A adds squeeze-and-excitation recalibration with global pooling and an MLP (25.1 M parameters; 89.0% balanced accuracy and 95.4% AUC-ROC). Branch B adds eight-head self-attention over 49 tokens, raising the count to 41.9 M (+67%) without a measurable gain (88.3% balanced accuracy).

Removing the 301 duplicates leaves aggregate results nearly unchanged: test accuracy 86.7% against 86.6% and AUC-ROC 94.8% against 95.0%. In the legacy test set ($N = 532$), 70 recordings are twins of training files. They scored 75.7% (Wilson interval 64.5% to 84.4%) against 88.3% (85.0% to 91.0%) on the 462 unique recordings. Twins come from different subsets than unique rows, which confounds the comparison. An earlier training run, before the batch-normalization and early-stopping fixes, scored 97.1% on twins against 89.8% on unique recordings, so the size of any inflation depends on the run. Duplicates remain a contamination to remove, but on this dataset they do not measurably change the reported scores.

B. Component Ablation

Table IV gives the five variants averaged over three seeds ($N = 486$ test recordings per seed).

Differences fall within seed and sampling variance. The baseline reaches $87.3 \pm 2.7\%$ accuracy ($88.4 \pm 2.4\%$ balanced accuracy). ResNet50+SE reaches $88.5 \pm 0.4\%$ ($89.0 \pm 1.8\%$) with AUC-ROC $95.4 \pm 0.3\%$. MHSA alone gives $86.6 \pm 1.6\%$ ($87.7 \pm 2.2\%$) and SE+MHSA gives $86.6 \pm 2.5\%$ ($88.3 \pm 2.5\%$). SE+MHSA beats the baseline in one of three seeds. With three seeds no significance test is possible, so we read the table as

TABLE IV
ARCHITECTURE COMPONENT ABLATION ON DEDUPLICATED BENCHMARK (MEAN OVER 3 SEEDS)

Model Variant	Accuracy (%)	Balanced Acc. (%)	Macro F1 (%)	AUC-ROC (%)	Sensitivity (%)	Specificity (%)	Params
ResNet50 (Baseline)	87.3	88.4	83.0	95.2	90.2	86.6	24.6M
+ SE Block	88.5	89.0	84.2	95.4	89.9	88.1	25.1M
+ MHSA (no SE)	86.6	87.7	82.1	95.1	89.6	85.9	41.3M
+ SE + MHSA [2]	86.6	88.3	82.2	95.0	91.2	85.4	41.9M
+ SE + MHSA (Dilated)	85.3	87.0	80.7	95.0	89.9	84.1	41.9M

TABLE V
LEAVE-ONE-DATABASE-OUT (LODO) CROSS-SITE EVALUATION (SEED 42)

	Baseline (ResNet50)		SE + MHSA [2]	
Held-out Subset	**Balanced Acc.**	**AUC-ROC**	**Balanced Acc.**	**AUC-ROC**
training-a	58.2%	60.6%	51.7%	49.1%
training-b	51.8%	62.6%	58.2%	62.5%
training-e	45.9%	27.7%	45.6%	19.9%
training-f	40.3%	48.0%	55.4%	57.2%
Mean Held-Out	49.1%	49.7%	52.7%	47.2%
In-distribution (dedup mean)	88.4%	95.2%	88.3%	95.0%
Drop vs. in-distribution	39.3% pts	45.5% pts	35.6% pts	47.8% pts

TABLE VI
FREQUENCY BAND OCCLUSION AUDIT WITH RANDOM CONTIGUOUS CONTROLS

Frequency Band	Masked ΔBA	Control ΔBA (Mean $\pm$ SD)	Reliance
B1 (20–150 Hz)	-7.81%	-2.01 $\pm$ 1.59%	Yes
B2 (150–500 Hz)	-2.41%	-2.83 $\pm$ 1.83%	No
B3 (500–1000 Hz)	-4.69%	-3.64 $\pm$ 3.30%	No

TABLE VII
EXPLANATION FAITHFULNESS: DELETION AND INSERTION CURVE AUC

Explanation Method	Deletion AUC	Random Del. AUC	Insertion AUC	Wi
Grad-CAM	0.629	0.527	0.673	
Attention (MHA)	0.735	0.525	0.640	
SHAP (Gradient)	0.567	0.526	0.531	

showing no measurable benefit from MHSA, not as proof that it is harmful. SE adds 0.5 M parameters to the baseline, while MHSA raises the count from 25.1 M to 41.9 M (+67%). The 7×7 grid supplies 49 tokens, a short sequence for long-range attention. The dilated variant scores $85.3 \pm 2.8\%$ accuracy ($87.0 \pm 2.4\%$ balanced), so a denser grid does not help under this training volume. SE is therefore the cheapest variant with no measurable loss.

C. Cross-Site Generalization

Table V gives LODO results for the four sources: MIT/Children's Hospital Boston (training-a), Aalborg University (training-b), Dalian University of Technology (training-e) and Soonchunhyang/Shiraz University (training-f) [20].

Every architecture degrades sharply. Mean held-out balanced accuracy is 49.1% for the baseline (a 39.3-point drop) and 52.7% for SE+MHSA (35.6 points), with AUC falling from 95.0% to 47.2% for the latter. On training-e, SE+MHSA reaches 45.6% balanced accuracy and 19.9% AUC. Prior shift alone cannot explain this, because AUC does not depend on class prior and a value below 0.5 means the ranking is inverted. The training pool mixes sources with very different abnormal rates (training-e is about 91% normal), so the network may have learned site signatures that correlate with labels. This hypothesis fits the data but was not tested directly.

D. Explanation Audit

The audit uses the SE+MHSA model (seed 42), because attention maps exist only for models with MHSA.

Occlusion. Masking B1 lowers balanced accuracy by 7.81%, beyond the control threshold ($-2.01\% \pm 1.59\%$, threshold -5.18%). Masking B2 (-2.41%) and B3 (-4.69%) stays within control variance. Keeping only B1 retains 83.66% balanced accuracy and 93.94% sensitivity. The model therefore relies on 20 to 150 Hz, but that band mixes S1 and S2 energy with stethoscope resonance and low-frequency room noise, so the result does not prove clinical reasoning.

Faithfulness. Deletion AUCs (0.629 Grad-CAM, 0.735 attention, 0.567 SHAP) all sit above random deletion (0.526) and the one-sided tests give $p \approx 1.00$. By this measure no method beats random removal. A constant zero-decibel baseline creates sharp artificial edges that the network never saw in training, which may distort its response. A baseline built from the mean spectrogram or a blurred copy would test this explanation.

Sanity check. Under cascading randomization the Grad-CAM correlation falls to -0.02, so the maps depend on learned weights [6].

Agreement. Pairwise Spearman correlations range from -0.12 to $+0.20$ and top-20% IoU stays below 0.21. Part of the gap is built in, since attention is class-agnostic and Grad-CAM targets one class. The three methods should therefore not be treated as confirming each other.

E. Calibration and Operating Point

For the SE model (seed 42) the uncalibrated ECE is 3.4%. Choosing $\tau = 0.61$ on validation probabilities for 90% validation sensitivity gives test sensitivity 86.9%, specificity 90.4% and balanced accuracy 88.7%. For the SE+MHSA model (seed 42) the uncalibrated ECE is 8.0%; choosing $\tau = 0.32$ gives test sensitivity 84.8%, specificity 85.5% and balanced accuracy 85.2%. In both architectures the 90% validation sensitivity

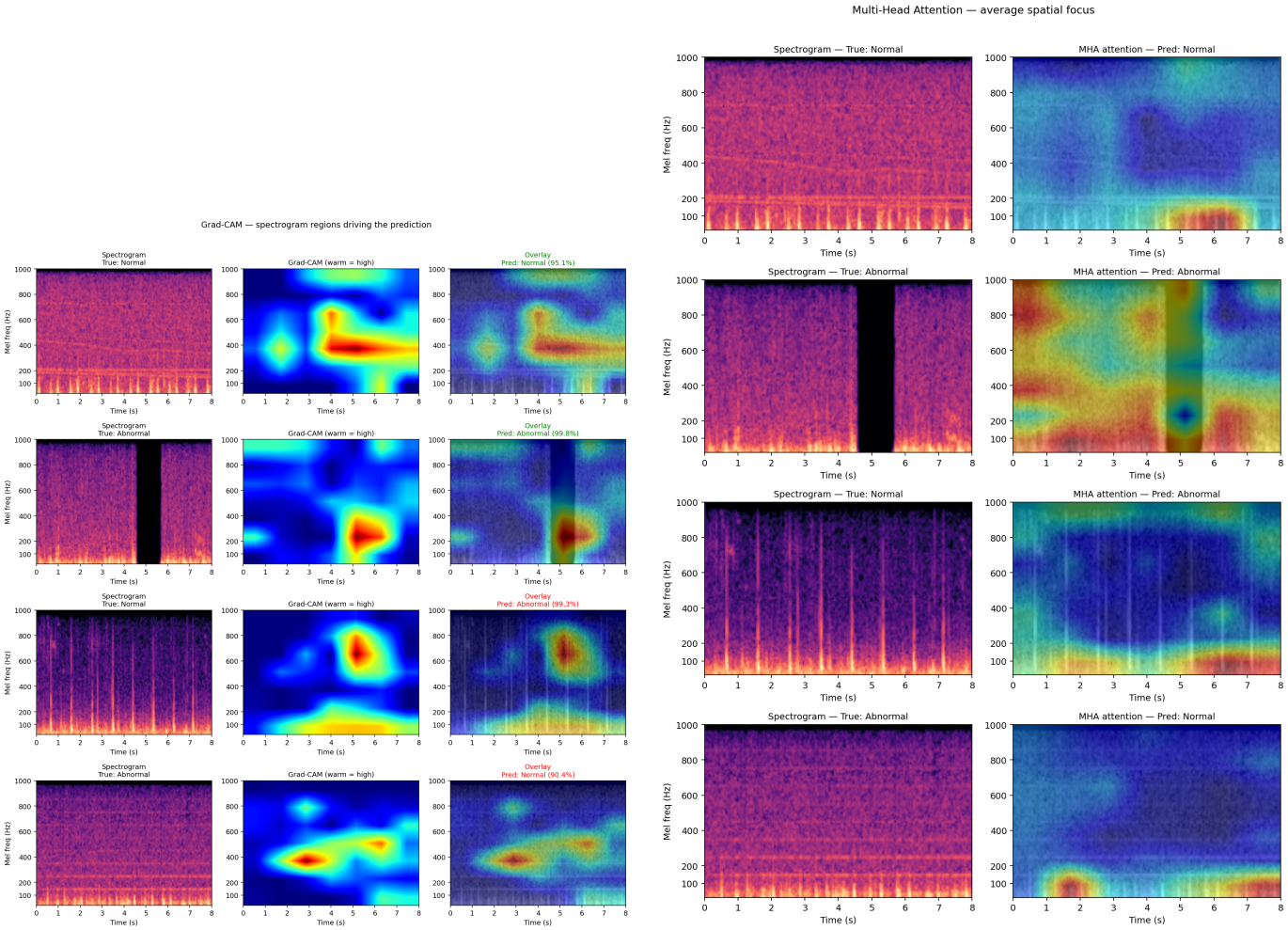


Fig. 2. Attribution maps for four test recordings (true normal, true abnormal, false positive and false negative). Left: Grad-CAM overlays on log-mel spectrograms. Right: head-averaged attention maps on the same recordings, which are spread widely across tokens.

TABLE VIII
MODEL CALIBRATION AND CLINICAL OPERATING POINT EVALUATION

Metric / Operational Parameter	**Value**
Expected Calibration Error (ECE, 15 bins)	8.0%
Validation-Selected Operating Threshold ($\geq$90% Sens.)	0.3
Test Sensitivity at Operating Threshold	84.8%
Test Specificity at Operating Threshold	85.5%
Test Balanced Accuracy at Operating Point	85.2%

target does not fully transfer to the test set, demonstrating an operational generalization gap in clinical threshold selection.

F. Per-Database Performance

Training-e scores 99.1% accuracy (99.7% normal and 92.6% abnormal), which a majority-class rule would nearly match given its class mix. Training-a and training-f reach 66.1% and 52.9% overall accuracy with clear sensitivity-specificity trade-offs. These gaps anticipate the LODO collapse.

TABLE IX
PER-DATABASE PERFORMANCE ON DEDUPLICATED TEST SPLIT (SE + MHSA, SEED 42)

Database	**Samples**	**Accuracy (%)**	**Normal Acc. (%)**	**Abnormal Acc. (%)**
training-a	62	66.1%	16.7%	86.4%
training-b	73	72.6%	77.6%	53.3%
training-c	5	80.0%	100.0%	75.0%
training-d	8	75.0%	50.0%	100.0%
training-e	321	99.1%	99.7%	92.6%
training-f	17	52.9%	66.7%	20.0%
Overall	**486**	**88.7%**	**91.0%**	**79.8%**

G. Comparison with Related Work

On PhysioNet 2016 our in-distribution accuracy (86.6% to 88.5%) is comparable to Li et al. [13] (86.80%) and below Althaph and Challa [16] (90.3%). Studies differ in partitioning, preprocessing and deduplication, so we make no ranking claim. Alrabie and Barnawi [5] used segmented HeartWave audio and Padhy et al. [15] used a balanced five-class subset, so their figures are not comparable. The contribution here is

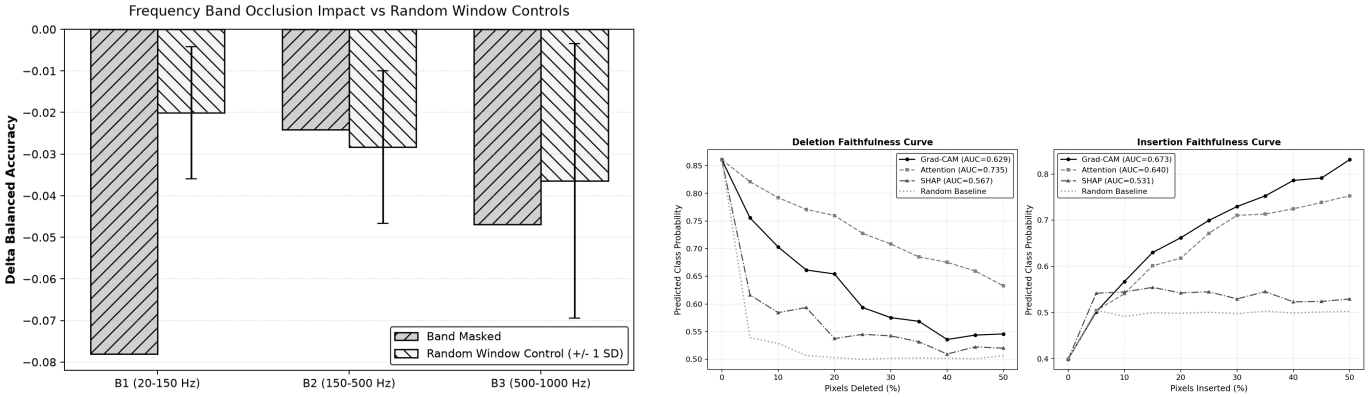


Fig. 3. Left: change in balanced accuracy when bands B1 to B3 are occluded, against 20 row-matched random controls ($\pm 1\sigma$). Only B1 exceeds the control threshold (-7.81% against -5.18%). Right: deletion and insertion curves for Grad-CAM, attention, SHAP and random attribution.

the evaluation protocol.

H. Limitations

Evaluation covers a single corpus without patient identifiers or longitudinal tracking. Three seeds allow no significance testing. Seeds change both the split and the initialization. The audit uses one seed and 128 test samples per analysis. The ResNet50 backbone was pretrained on natural images. The 7×7 grid limits explanations to 1.14 s per cell; the dilated grid of 0.57 s did not improve accuracy. Grad-CAM was not scored against clinician annotations. State enrichment against Springer segmentations was not computed because no annotations were found in our copy of the data. Cropping to 8 s discards later cardiac cycles.

VI. Conclusion and Future Work

We audited an explainable heart sound classifier on PhysioNet/CinC 2016 along five lines. Duplicate validation files contaminate the corpus but did not measurably change scores. Self-attention added 16.8 M parameters without a measurable benefit; ResNet50 with SE matched or exceeded every larger variant at 25.1 M parameters. Leave-one-database-out testing cut mean balanced accuracy to about 49% to 53%, so single-split scores overstate real-world performance. Grad-CAM depends on learned weights and the model relies on the 20 to 150 Hz band, yet no attribution method passed deletion faithfulness and the three methods agree weakly.

Future work will use patient-grouped and site-stratified partitions, report more seeds with confidence intervals and test whether site-invariant training or sensor normalization lifts LODO scores. We will repeat the deletion test with in-distribution baselines, compute state enrichment from segmentations and score Grad-CAM against clinician annotations.

Code Availability

The implementation, preprocessing scripts, checkpoints, audit suites and evaluation scripts are at https://github.com/urvagandhi/heart-sound-analysis.

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